

FRANCIS SCOTT KEY BRIDGE, MD, USA

WMO: 997302

Lat: 39.218N

Lon: 76.527W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -6.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 15.4	(k) -0.6	(l) 14.4	(m) -1.0	(n) 7.0	(o) 290	(p) 0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.5	(c) 32.4	(d) N/A	(e) 31.1	(f) N/A	(g) 29.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.0	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.4	(b) 12.0	(c) 10.5	(e) -10.7	(f) 34.8	(g) 2.3	(h) 1.5	(i) -12.3	(j) 35.9	(k) -13.7	(l) 36.7	(m) -14.9	(n) 37.5	(o) -16.6	(p) 38.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	1.8	2.5	6.8	13.5	18.6	23.5	26.2	25.2	21.4	15.4	9.2	5.0
	DBStd	9.39	4.99	4.76	4.74	4.35	3.77	2.93	2.31	2.26	3.02	4.15	4.04	4.25
	HDD10.0	841	256	214	123	15	0	0	0	0	6	62	165	
	HDD18.3	2318	512	443	358	156	45	2	0	0	7	109	274	413
	CDD10.0	2356	3	4	24	119	265	405	501	470	343	174	38	9
	CDD18.3	792	0	0	1	10	52	157	243	212	100	18	0	0
	CDH23.3	5905	0	1	3	71	239	1080	2289	1674	483	64	0	0
	CDH26.7	1587	0	0	0	18	39	271	742	425	79	13	0	0
(14) Wind	WSAvg	4.7	4.9	5.2	5.4	5.2	4.9	4.8	4.1	4.0	4.1	4.6	4.6	4.6
(15) Precipitation	PrecAvg	1170	83	70	109	92	102	112	106	112	113	99	76	95
	PrecMax	1680	149	163	224	181	212	293	178	283	307	222	153	183
	PrecMin	845	36	14	7	49	32	41	25	38	3	8	21	40
	PrecStd	181	26	37	46	35	44	62	40	53	78	60	38	39
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	20.1	23.0	29.1	30.2	32.8	35.0	33.5	31.0	28.7	21.0	19.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	13.9	14.8	19.0	25.3	27.7	31.1	33.0	31.4	28.7	24.9	18.3	15.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	11.1	11.3	16.1	22.2	25.9	29.5	31.6	30.1	27.4	23.0	16.7	13.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	8.8	8.6	13.6	19.9	24.4	28.1	30.3	29.0	26.1	21.5	15.2	11.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.5	7.0	7.6	6.9	6.4	6.5	5.8	5.9	6.4	6.2	5.7
		MCDBR	10.0	10.9	11.2	10.9	8.2	7.9	8.0	7.1	7.0	8.0	8.2	9.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.311	0.321	0.362	0.400	0.446	0.499	0.518	0.513	0.428	0.363	0.326	0.311
	taud		2.470	2.427	2.375	2.290	2.196	2.090	2.053	2.051	2.260	2.446	2.510	2.511
	Ebn at Noon		862	902	893	876	839	792	773	764	813	841	838	836
	Edh at Noon		79	95	111	129	144	161	165	162	123	92	76	72
(44) All-Sky Solar Radiation	RadAvg		2.04	2.89	3.83	4.92	5.51	6.11	6.05	5.35	4.36	3.23	2.33	1.79
	RadStd		0.17	0.33	0.29	0.40	0.66	0.47	0.29	0.30	0.51	0.35	0.20	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	+73

Nomenclature: See separate page